

Breathing Under Pressure

How Deforestation, Urbanization & Environmental Spending Impact Air Quality and Public Health Across Global Cities

Sector

Environmental Analytics

Team ID

G - 10

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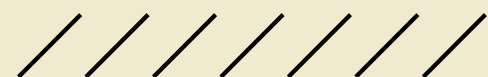
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Why This Problem Matters

Over 90% of the world population breathes polluted air. Cities are expanding rapidly, forests are shrinking, and healthcare burdens are rising.

Objective

Enable smarter environmental investment and urban planning decisions.

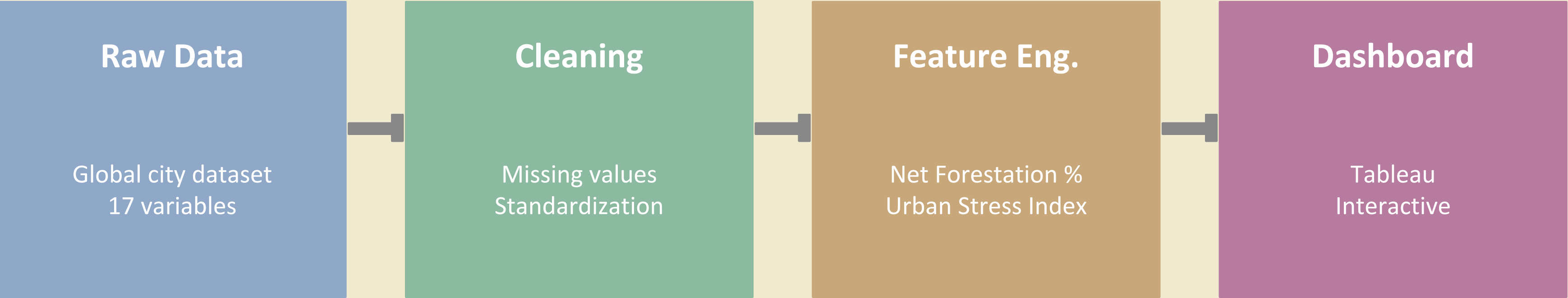
Governments must decide:

- Where to invest environmental budgets
- Which cities need urgent intervention
- Which strategies actually improve public health



Data Pipeline

Global city-level dataset · 17 variables · Multi-year time series



Feature Engineering Highlights



Net Forestation %

Forest gain – forest loss per region



Urban Stress Index

Composite density + vehicle + AQI score

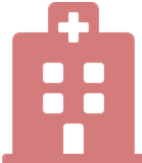


Sustainability Score

Budget efficiency × green space ratio

KPI & Metrics Framework

KPIs represent three layers: Drivers → Interventions → Outcomes

DRIVERS	INTERVENTIONS	OUTCOMES
AQI Index Air Quality Index per city	Net Forestation % Forest gain minus forest loss	Life Expectancy Average years at birth
PM2.5 Level Fine particulate matter (µg/m³)	Environmental Budget Govt. spending on env. programs	Health Cost Index Healthcare burden per capita
CO₂ Emissions Carbon output per capita		
Vehicle Growth Rate Annual % increase in vehicles		

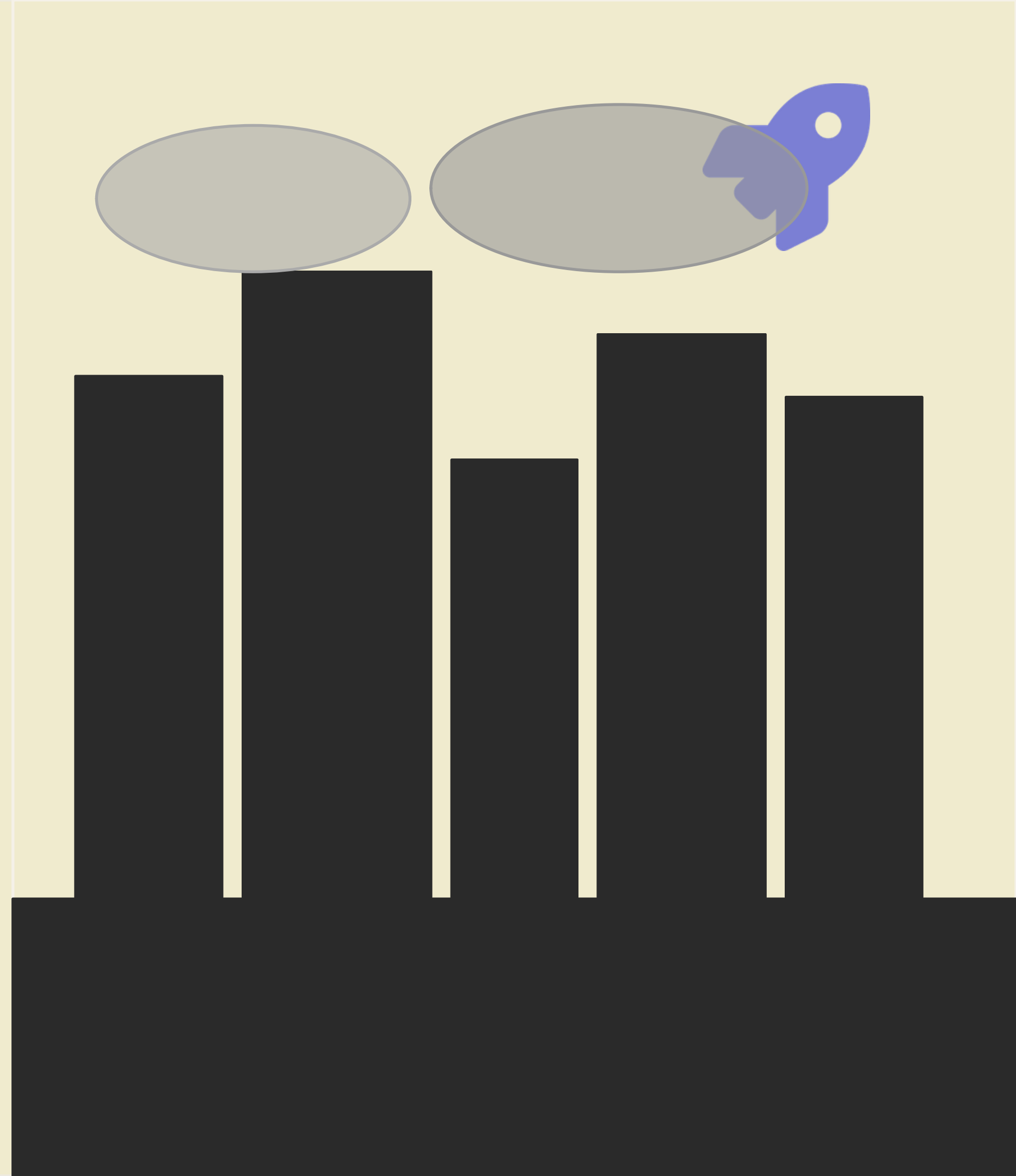
★ Vehicle Growth Rate is the strongest single predictor of rising AQI across all cities studied

Key Insights (EDA)

Pollution is driven by controllable urban planning decisions.

Major Findings

- Higher AQI cities show lower life expectancy
- Vehicle growth strongly increases pollution
- More green space = cleaner air
- Forest loss worsens AQI trends
- Dense cities face higher urban stress
- Strategic environmental spending improves outcomes



Advanced Analysis

Correlation Analysis

AQI rises with vehicle density, CO₂ output, and city population density

Trend Analysis

AQI improves where green space % and env. budget both increase over time

City Segmentation

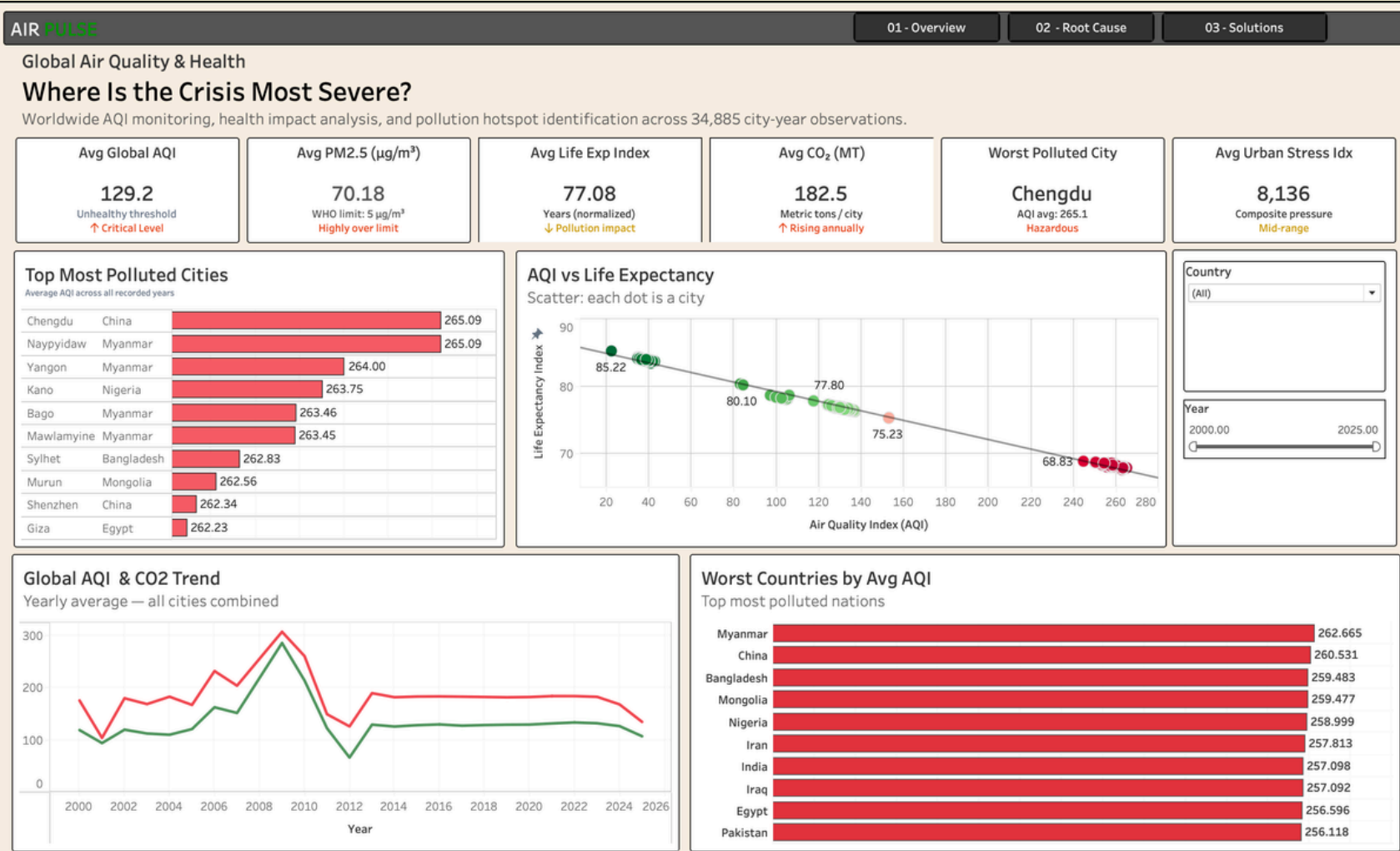
Cities clustered into High-Risk vs Model Cities based on composite AQI score

City Segment Comparison

Metric	High-Risk Cities	Model Cities
AQI Level	180 – 250+	30 – 60
Green Space	< 5%	> 20%
Vehicle Growth	> 8% / yr	< 3% / yr
Env. Budget	Low allocation	High allocation
Life Expectancy	Significantly lower	Significantly higher

KEY FINDING: AQI is 4× worse in High-Risk cities — but the gap closes with targeted investment

Interactive Dashboard



The Impact We Can Create

INACTION



2M+ preventable deaths/yr



Healthcare costs surge



AQI above 200 in cities



Forest cover continues falling

STRATEGIC ACTION



15–25% AQI reduction



Healthier, longer lives



3× ROI on env. investment

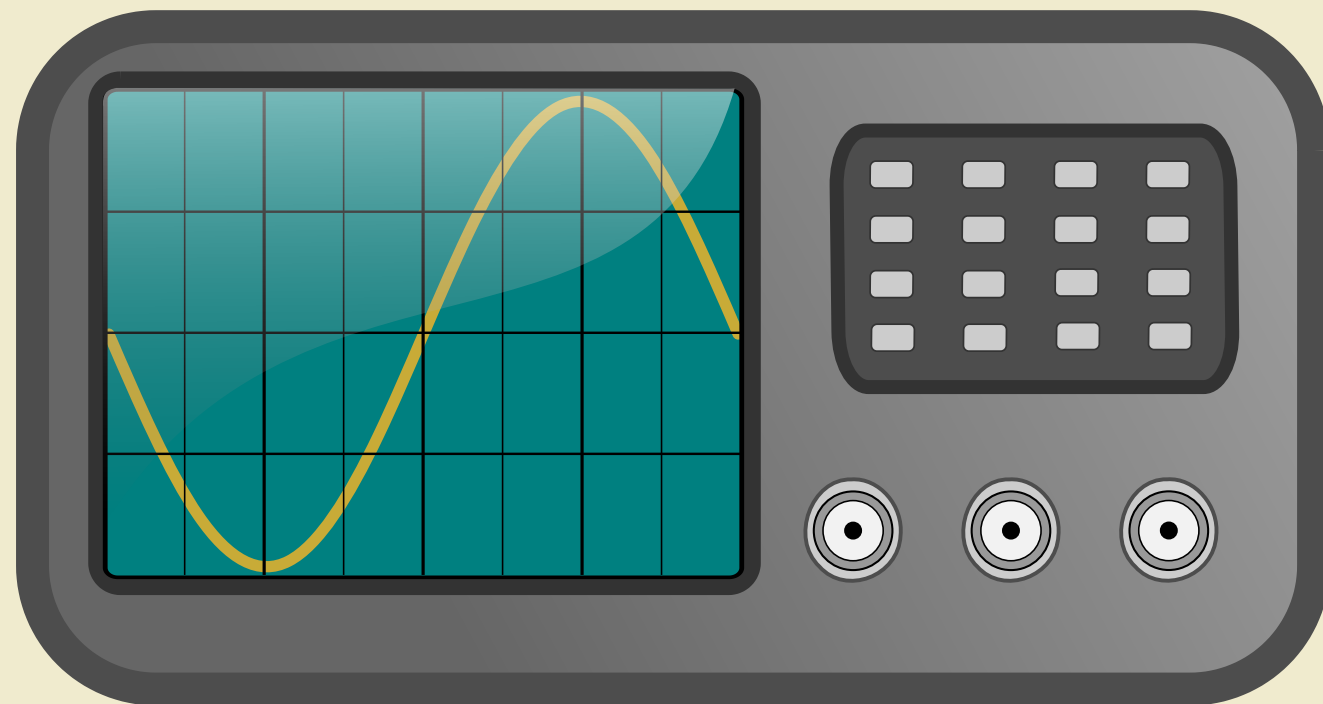


Restored forest ecosystems

Limitations & Next Steps

Current Limitations

Data may lag real-time conditions by weeks or months
Some cities have incomplete or inconsistently reported records
Assumes consistent AQI methodology across all countries
Correlations identified — causal claims require further study
Budget data varies in reporting standards across governments



Roadmap

01

Predictive ML Model

Forecast city AQI under different policy scenarios

02

Real-Time Dashboard

Live AQI feed integrated with government open APIs

03

Policy Simulator

'What-if' tool — adjust budget, vehicles, green space

04

Global Scale-Up

Expand dataset to 500+ cities across all continents

Recommendations

Insight → Action → Impact



Green Infrastructure

Add urban parks & tree corridors

Impact: AQI ↓ 15–25%



Reduce Vehicle Emissions

EV incentives & congestion pricing

Impact: PM2.5 ↓ 20%



Optimize Budget Use

Prioritize highest-AQI cities first

Impact: 3× ROI on health



Protect Forests

Halt deforestation in risk zones

Impact: Long-term AQI stability



Focus High-Risk Cities

Tier cities by AQI + health burden

Impact: 2M+ cases prevented/yr



Data-Led Governance

Real-time AQI dashboard for officials

Impact: Faster policy response